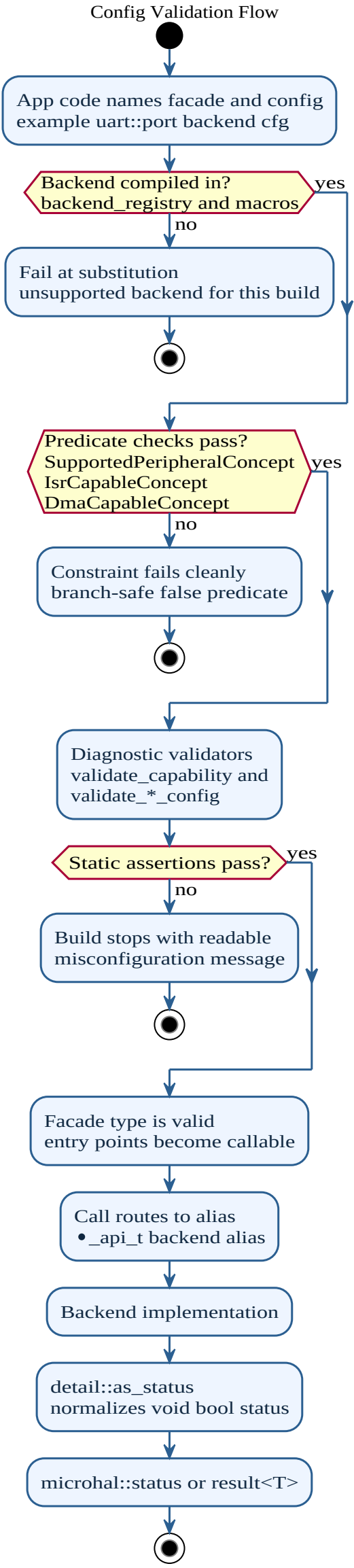
Config Validation Flow
App code names facade and config
example uart::port backend cfg
Backend compiled in?
backend_registry and macros
yes
no
Fail at substitution
unsupported backend for this build
Predicate checks pass?
SupportedPeripheralConcept
IsrCapableConcept
DmaCapableConcept
yes
no
Constraint fails cleanly
branch-safe false predicate
Diagnostic validators
validate_capability and
validate_*_config
Static assertions pass?
yes
no
Build stops with readable
misconfiguration message
Facade type is valid
entry points become callable
Call routes to alias
• _api_t backend alias
Backend implementation
detail::as_status
normalizes void bool status
microhal::status or result<T>